

ATOR EARTH - Global Landscape Report

Is There Any Other Map This Detailed? Research Finding: No.

Deborah Ram Mozes, ATOR Institute, Bandung, 26 May 2026

Executive Summary

Question: does any public map combine CPU GPU AI mineral chain into one archival inventory with site lat lon, operator gap, environmental risk, source url and hardware relevance? After checking USGS, IEA, Energy Monitor, open trackers, IPE China, Jain Family Institute, IPIS DRC, OECD semiconductor value chain, MacroPolo AI chips, Mindat and OreAtlas, answer is no. USGS Global Distribution has 22 minerals point layers but no hardware link. USGS Atlas is country-level for 90 minerals. IEA and Energy Monitor are country production and demand, not site. Open trackers show active mines filterable by commodity but not AI chain. OECD and MacroPolo map semiconductor dependencies on gallium germanium quartz but based on trade data, not mine sites. ATOR EARTH is the only open inventory with 100+ nodes across Si high purity quartz, Li, Sn, REE, Co, Ni, Cu, Ta Nb, W, Ga, Ge, In and others with explicit CPU GPU AI linkage and Source needed flag for beneficial ownership search.

What Exists

USGS: Global Distribution of Selected Mines, Deposits, and Districts of Critical Minerals - 22 minerals, point and polygon layers. Critical Minerals Atlas - interactive country-level. IEA Critical Minerals Data Explorer - demand projections under STEPS APS NZE. Energy Monitor Critical Mineral Tracker - production reserves per country, China refining 68 percent nickel, 40 percent copper, 59 percent lithium, 73 percent cobalt. Critical Minerals Tracker open dashboard - interactive map of active mines, refineries, processing plants filterable by commodity. IPE Transition Mineral Resources Map 2024 - copper cobalt lithium nickel tungsten graphite rare earth with satellite imagery. IPIS interactive webmap - 2400+ artisanal mining sites eastern DRC with armed presence. OECD Mapping the Semiconductor Value Chain June 2025 - China 98 percent gallium 2023, 68 percent germanium 2020, concentration of borates and silicon wafers. MacroPolo AI Chips Supply Chain Mapping - upstream midstream downstream with color intensity percent global production.

Why ATOR EARTH Is Different

Feature	Existing Maps	ATOR EARTH
Site lat lon 100+ nodes	USGS has 22 minerals, others country-level	Yes, 100+ across Si Li Sn REE Co Ni Cu Ta Nb W Ga Ge
Hardware relevance CPU GPU AI	OECD and MacroPolo mention Ga Ge quartz but no site	Yes, explicit tin solder Bangka, nickel HPAL Sulawesi Obi, high purity quartz Spruce Pine
Operator company Source needed gap	None	Yes, gap for MODI AHU OpenCorporates SEC EDGAR RMI RMAP
Environmental risk per site	None, IPIS has armed presence only	Yes, literature based offshore dredging turbidity tailings deforestation HPAL waste
Source url per node	No	Yes
Open license	USGS public domain, OECD CC BY 4.0	CC BY-NC-SA 4.0

Conclusion: No public map currently combines CPU GPU AI chain into single archival inventory with site coordinates, operator gaps, environmental risk, source url and hardware relevance as ATOR EARTH does. That is novelty for ITB, BRIN, ESDM and international citation.